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[22BSBE01]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER - 2023

BIOLOGY FOR ENGINEERS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ii. Answer all questions.

		PART-A	**05 X 02=10**		
1.	**a.**	Define pharmacology.	L-1	CO1	**02**
	b.	What are lithotrophes? Give two examples.	L-1	CO2	**02**
	c.	Write the Michaelis-Menten equation and describe the terms	L-2	CO3	**02**
	d.	Draw the energy profile diagram for endergonic and exergonic reactions.	L-2	CO4	**02**
	e.	Describe growth kinetics.	L-2	CO5	**02**
		PART-B	**06 X 05=30**		
2.	**a.**	Summarize the differences between prokaryotic and eukaryotic cells.	L-2	CO1	**05**
		OR			
	b.	Briefly discuss three types of RNAs and their biological roles	L-2	CO1	**05**
3.	**a.**	What does kingdom Monera consist of? Write its four types based on the shapes with suitable examples	L-2	CO2	**05**
		OR			
	b.	Define autotrophs. Mention the two types of autotrophs with examples.	L-2	CO2	**05**
4.	**a.**	Outline the various enzymes involved in digestion process.	L-2	CO3	**05**
		OR			
	b.	Explain the four methods used to monitor the enzyme catalyzed reactions.	L-2	CO3	**05**
5.	**a.**	Discuss how the first law of thermodynamics related in biology.	L-2	CO4	**05**
		OR			
	b.	Explain the cellular respiration process.	L-2	CO4	**05**
6.	**a.**	Elaborate the nutrition, reproduction, and excretion in single cellular organisms.	L-2	CO5	**05**
		OR			
	b.	Discuss any five methods used for the identification of microorganisms.	L-2	CO5	**05**
7.	**a.**	Describe sterilization. Write three types of sterilization methods.	L-2	CO5	**05**
		OR			
	b.	Explain the essential components of media composition (culture media).	L-2	CO5	**05**

		PART-C		**06 X 10=60**	
8.	**a.**	i. Summarize the salient features of DNA ii. Discuss five important structural components present in cytoplasm of eukaryotic cells	L-2	CO1	10
		OR			
	b.	i. Define pharmacokinetics and its four phases (ADME) ii. Detail the concept of chemotherapy and discuss its four types	L-2	CO1	10
9.	**a.**	Explain the classification of organisms based on energy utilization and excretion.	L-2	CO2	10
		OR			
	b.	Write the characteristic features of kingdom animalia and kingdom plantae.	L-1	CO2	10
10.	**a.**	Narrate the five stages involved in enzyme catalyzed reactions with a neat energy profile diagram.	L-2	CO3	10
		OR			
	b.	Explain the DNA replication process with the detailed role of each enzyme involved.	L-2	CO3	10
11.	**a.**	Draw the chemical structure of ATP and explain why it is referred as the energy currency.	L-2	CO4	10
		OR			
	b.	Write the salient features of energy yielding and energy consuming reactions with examples.	L-1	CO4	10
12.	**a.**	i. Explain the biological species concept and its limitations ii. Classify microorganism based on taxonomy, morphology, physiology, and ecological roles.	L-2	CO5, CO6	**10**
		OR			
	b.	i. What is a microscope? Discuss the three types of microscopes. ii. Interpret the ecological aspects of single celled organisms.	L-2	CO5	**10**
13.	**a.**	i. Write the various levels involved in protein structure with neat diagrams. ii. Write a short note on central dogma of life.	L-1	CO1	**10**
		OR			
	b.	i. Describe vaccines. Discuss the three important types of vaccines. ii. Define half-life of a drug. Write the formulae for its calculation.	L-2	CO1	**10**